REFERENCE · CHAPTER 13

End-to-End Audit Attribution (W3C Trace, BigQuery Sink) & Kill Switch Runbook

Linking every SQL query to the human caller and agent turn, enforcing Principal Access Boundaries (PAB), and executing the 3-step emergency kill switch.

1. Closing the Audit Blind Spot (`T8`)

By default, Cloud SQL, AlloyDB, and Spanner disable `DATA_READ` and `DATA_WRITE` audit logs, and logs in `_Default` expire after 30 days. Our control plane enables Data Access logs across all data stores, routes them to a 400-day partitioned BigQuery sink, propagates W3C `traceparent` from the client edge into SQL comments/job labels, and streams full agent trajectories via ADK's `BigQueryAgentAnalyticsPlugin`.

2. The 3-Step Emergency Kill Switch Runbook

STEP	ACTION	COMMAND / CONTROL PLANE TARGET	FORENSIC IMPACT
Step 1: Block Agent Principal	Apply an immediate IAM Deny policy & Principal Access Boundary (PAB) ceiling on the agent's SPIFFE identity, and revoke `roles/iap.egressor` at the Egress Gateway.	`gcloud iam deny-policies create` / `IAM PAB`	Halts all tool and database calls within seconds while keeping the container intact.
Step 2: Revoke Delegated Tokens	Revoke active OAuth refresh tokens and OBO grants in `Auth Manager` and rotate the `ALLOWED_KIDS` KMS context signing key version if needed.	`Auth Manager` / `Cloud KMS`	Invalidates any in-flight tokens immediately across all MCP servers.
Step 3: Preserve Evidence	Snapshot Cloud Run / Agent Runtime revisions, export `Memory Bank` sessions, and query the 400-day BigQuery audit sink by `trace_id` and `spiffe_id`.	`BigQuery Audit Sink` + `Security Command Center`	Preserves complete chain of custody for incident response.